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(1) Research Program/Project/Study Title: Development of Shrimp (<i>Metapenaeus ensis</i>) Ice Cream	
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(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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ABSTRACT

The composition and evaluation of a Shrimp Ice Cream are composed of the following ingredients: shrimp, the main ingredient that is locally available and rich in nutritional factors. Arrowroot is added to serve as a stabilizer and binder of the product. Fresh milk and coconut milk were added to provide creaminess and richness to ice cream and contributes to its melting characteristics—salt and sugar to enhance its flavor. The experimental development research generally aimed to provide an innovative and unique ice cream product that increases the production of greasy back shrimp and a new taste. Specifically, it sought to (1) Determine the level of acceptability of Shrimp Ice Cream: (2) Determine the sensory properties of the Shrimp Ice Cream in terms of taste, texture, aroma, color, and general acceptability; and (3) Determine the consumers' acceptability of Shrimp Ice Cream. Analysis of variance in Completely Randomized Design (CRD) was used to analyze the data, and a significant difference occurred. In the four Shrimp ice cream formulations, it was found that Treatment 3 (15g of shrimp broth and 10g shrimp bits) was the most acceptable product as perceived by the consumers. The product was described as "Like very much shrimp flavor" in terms of taste," Like very much Creamy texture "in terms of texture, "like very much" when it comes to aroma, and "Off White" in terms of color. The Hedonic test results highlighted that treatment 3 (15g of shrimp broth and 10g shrimp bits) had the best result in terms of taste, texture, aroma and color, treatment 3 (15g of shrimp broth and 10g shrimp bits) also had the highest general acceptability. There are significant differences existing among the four treatments of Shrimp Ice cream.

Keywords;

Shrimp, ice cream, Seafood ice cream.

INTRODUCTION

Ice cream is the most known dairy dessert, it is considered a nutritive food because it is mainly made of milk and fruits, representing a source of proteins, some vitamins, minerals, and some phytochemicals (*Balthazar, C. F. et al.*)

The greasyback shrimp (*Metapenaeus ensis*) is a commercially important, brackish-water crustacean species (Chu et al., 1995), widely distributed in many Asian countries (Chu and So, 1987). *Metapenaeus ensis* is a common local shrimp of commercial importance especially in Panay Island. The nutritional profile of Shrimp is excellent. Shrimp is a good source of protein and is also high in a variety of vitamins and minerals. Since prawns include omega-3 fatty acids and the antioxidant astaxanthin, eating prawns may also be good for your heart and brain

Ice cream is a dairy frozen dessert made from milk, cream, sugar, and additive flavorings such as chocolates, fruits, and other ingredients added to ice cream formulation. Ice cream has elements that help strengthen your immune system. Milk contains Vitamins and nutrients including Vitamin A, Vitamin D, selenium, and zinc which are all essential for a healthy immune system. Ice cream can also make us happier and can satisfy pregnancy cravings.

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Thus, this study aims to provide an innovative and unique ice cream product that increases the production of greasy back shrimp and a new taste, combining the sweetness of ice cream with the savory and umami flavors everyone can enjoy.

OBJECTIVES

This study aims to;

1. Determine the level of acceptability of Shrimp Ice Cream;
2. Determine the sensory properties of the Shrimp Ice Cream in terms of taste, texture, aroma, color, and general acceptability; and
3. Determine the consumers' acceptability of Shrimp Ice Cream.

METHODOLOGY

The study used developmental research at the Fisheries and Food Research Development Center at Capiz State University, Dayao Satellite College, Dayao, Roxas, Capiz. The experiment was used to investigate the acceptability of Shrimp as a flavor of Ice cream.

Completely Randomized Design (CRD) was utilized in the study, where the different amount of Shrimp broth and Shrimp bits used to make Shrimp Ice Cream.

The Shrimp Ice Cream was carried out four (4) treatments. After processing, the samples was subjected to sensory evaluation and the result are are will evaluated using SPSS.

Table 1: Ingredients and measurements used in Shrimp Ice cream.

Ingredients	Measurements			
	Treatment 1	Treatment 2	Treatment 3	Treatment 4
Shrimp broth	5g	10g	15g	20g
Shrimp (bits)	20g	15g	10g	5g
Coconut Milk	1500ml	1500ml	1500ml	1500ml
Sugar	300g	300g	300g	300g
Arrowroot	150g	150g	150g	150g
Fresh Milk	1000ml	1000ml	1000ml	1000ml
Salt	5g	5g	5g	5g

Scoring was a 9-point hedonic scale for the sensory characteristics evaluated by 10 evaluators—a 9-point hedonic scale for consumers' preferences and 100 respondents.

The data was gathered after every replication and was collected, tallied, tabulated, and computed using mean for acceptability and Analysis of Variance (ANOVA) for the significant difference among treatments.

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Methods

Process for minced Shrimp

1. Clean the shrimp and remove the shell. Then, set aside the shrimp meat.
2. Add the shrimp shells and water into a blender and blend.
3. Add shrimp shells blended to a pan, then add water and salt. Boil for 15 minutes in a low heat. Strain and set aside the shrimp broth.
4. Chop the shrimp meat into mince and cook on low heat until it becomes flakey.

Ice Cream mixture:

1. Boil water and add the arrowroot. Stir until the mixture is sticky.
2. Combine fresh milk, coconut milk, sugar, shrimp broth, and arrowroot mixture in an ice cream tube.

Process for Ice cream:

1. Put the ice cream tube inside the ice cream container and add the ice-rock salt mixture around it.
2. Mix the ice cream mixture using a hand-held mixer until it becomes a smooth texture.
3. Add the minced shrimp and mix again.
4. Secure the ice-rock salt mixture and leave it overnight.

RESULTS AND DISCUSSION

In terms of taste, Table 2.0 shown that treatment 3 got the highest mean of 8.6, equivalent to the verbal interpretation of Like very much shrimp flavor. In comparison, treatment 4 got the lowest mean of 5.7, which equivalent to the verbal interpretation of Neither like nor dislike.

Table 2.0. Sensory characteristics of different treatments in terms of taste.

Treatments	Replication			Total	Mean	Verbal Interpretation
	1	2	3			
T1	7.1	7.1	6.1	20.3	6.8	Like moderate shrimp flavor
T2	7	5.6	5.8	18.4	6.1	Like slightly shrimp flavor
T3	8.6	8.5	8.7	25.8	8.6	Like very much shrimp flavor
T4	6.1	5.8	5.2	17.1	5.7	Neither like nor dislike

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In terms of texture, table 2.1 shown, that Treatment 3 got the highest mean of 8.4, which is equivalent to the verbal interpretation of Like very much Creamy texture. In comparison, treatment 2 (5.43) and treatment 4 (5.36) got the lowest mean, equivalent to the verbal interpretation of Neither like nor dislike.

Table 2.1. Sensory characteristics of different treatments in terms of texture

Treatments	Replication			Total	Mean	Verbal Interpretation
	1	2	3			
T1	7.3	7.3	6.5	21.1	7.03	Like moderate Creamy texture
T2	5.4	5.5	5.4	16.3	5.43	Neither like nor dislike
T3	8.4	8.5	8.3	25.2	8.4	Like very much Creamy texture
T4	5.1	5.5	5.5	16.1	5.36	Neither like nor dislike

In terms of aroma, Table 2.2 shown, treatment Treatment 3 got the highest mean of 8.6, which is equivalent to the verbal interpretation of Like Very Much. In comparison, Treatment 2 with a mean of 6.8, got the lowest score equivalent to the verbal interpretation of Like Moderately.

Table 2.2. Sensory characteristics of different treatments in terms of Aroma

Treatments	Replication			Total	Mean	Verbal Interpretation
	1	2	3			
T1	7.3	6.8	6.1	20.2	6.73	Like Moderately
T2	7.3	6	7.1	20.4	6.8	Like Moderately
T3	8.5	8.7	8.6	25.8	8.6	Like Very Much
T4	5.5	7.1	7	19.6	6.53	Like Moderately

In terms the color of shrimp ice cream, table 2.3 shown, Treatment 3 (has a mean of 8.5, which is equivalent to the verbal interpretation of Off White. In comparison, Treatment 1, 2 and 3 got the lowest mean results which is equivalent to the verbal interpretation of light cream.

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Table 2.3. Sensory characteristics of different treatments in terms of Color

Treatments	Replication			Total	Mean	Verbal Interpretation
	1	2	3			
T1	6.8	7.4	6.5	20.7	6.9	Light cream
T2	7.2	6.6	6.6	20.4	6.8	Light cream
T3	8.3	8.5	8.7	25.5	8.5	Off White
T4	6.6	6.3	6.8	19.7	6.56	Light cream

For the General Acceptability of shrimp ice cream, Table 2.4 shown , Treatment 3 got the highest mean of 8.56 which is equivalent to verbal interpretation of like extremely, while Treatment A with a mean of 6.2 which is equivalent to verbal interpretation of which is equivalent to Like Slightly.

Table 2.4. Sensory characteristics of different treatments in terms of General Acceptability

Treatments	Replication			Total	Mean	Verbal Interpretation
	1	2	3			
T1	6.2	6.2	6.2	18.6	6.2	Like Slightly
T2	7	5.9	6.3	19.2	6.4	Like Slightly
T3	8.6	8.6	8.5	25.7	8.56	Like Extremely
T4	6.2	6.7	6.2	19.1	6.36	Like Moderately

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RESULT OF SENSORY CHARACTERISTICS AMONG TREATMENTS

In Table 3.0. Using statistical analysis, results reveal that there is a significant difference in the scientific ability in the sensory characteristics in terms of taste.

Table 3.0 One-way ANOVA results of the sensory characteristics in terms of taste

	Sum of Squares	df	Mean Square	F	Sig.
Between groups:	32.200	3	10.733	12.795	.000
With groups:	30.200	36	.839	Permutation p (n=99999)	
Total:	62.400	39			

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

In Table 3.1. Using statistical analysis, results reveal that there is a significant difference in the scientific ability in the sensory characteristics in terms of texture.

Table 3.1 One-way ANOVA results of the sensory characteristics in terms of texture.

	Sum of Squares	df	Mean Square	F	Sig.
Between groups:	56.500	3	18.833	19.153	.000
With groups:	35.400	36	.983	Permutation p (n=99999)	
Total:	91.900	39			

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

In Table 3.2. Using statistical analysis, results reveal that there is a significant difference in the scientific ability in the sensory characteristics in terms of aroma.

Table 3.2 One-way ANOVA results of the sensory characteristics in terms of aroma.

	Sum of Squares	df	Mean Square	F	Sig.
Between groups:	67.275	3	22.425	46.131	.000
With groups:	17.500	36	.486	Permutation p (n=99999)	
Total:	84.775	39			

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

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In Table 3.3. Using statistical analysis, results reveal that there is a significant difference in the scientific ability in the sensory characteristics in terms of color.

Table 3.3 One-way ANOVA results of the sensory characteristics in terms of color.

	Sum of Squares	df	Mean Square	F	Sig.
Between groups:	17.300	3	5.767	6.070	0.02
With groups:	34.200	36	.950	Permutation p (n=99999)	
Total:	51.500	39			

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

In Table 3.4. Using statistical analysis, results reveal that there is a significant difference in the scientific ability in the sensory characteristics in terms of general acceptability.

Table 3.4 One-way ANOVA results of the sensory characteristics in terms of General Acceptability.

	Sum of Squares	df	Mean Square	F	Sig
Between groups:	38.700	3	12.900	27.000	.000
With groups:	17.200	36	.478	Permutation p (n=99999)	
Total:	55.900	39			

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

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CONSUMER ANALYSIS (ANOVA) RESULT

As the presented on the Table 4.0, the shrimp ice cream are most acceptable in college respondents and have a higher mean result of 8.88 which equivalent to the verbal interpretation of Like extremely.

Table 4.0. The mean consumer acceptability using 9-point hedonic scale

Respondents	Total	Mean	Verbal Interpretation
Elementary Students	208	8.32	Like extremely
High School Students	221	8.84	Like extremely
College Students	222	8.88	Like extremely
Employee	215	8.6	Like very much

**Legend: 1= Dislike Extremely, 2= Dislike Very Much, 3= Dislike Moderately, 4= Dislike Slightly, 5= Neither Like nor Dislike, 6= Like Slightly, 7= Like Moderately, 8= Like Very Much, 9= Like Extremely*

In Table 4.1, using statistical analysis, and one-way ANOVA, results shown that there is no significant difference in the consumer acceptability between four group respondents.

Table 4.1 One-way ANOVA results of the sensory characteristics in terms consumer acceptability.

	Sum of Squares	df	Mean Square	F	Sig.
Between groups:	.680	3	.227	.476	.699
With groups:	45.680	96	.476	Permutation p (n=99999)	
Total:	46.360	99			

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

CONCLUSION

- The result of the ice cream research is acceptable, with a general rating of like very much.
- In all sensory parameters Treatment 3 are acceptable with a rating of like very much.
- College respondents have higher scores on the consumer acceptability tests, interpreted as Like extremely.
- The shrimp ice cream result have potential for commercialization.

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RECOMMENDATIONS

- It can be recommended that other researchers develop a new dessert product using seafood.
- Conduct microbial analysis to determine the microbial content of shrimp ice cream.
- Conduct Shelf-life stability test for shrimp ice cream.
- Sensory characteristics of shrimp ice cream can still be enhanced by modification processes or by adding flavor or ingredients.
- This study is recommended for consumers as a dessert but not for those who have an allergy to shrimp.

Output (Indicate the specific products that the project has produced)

Shrimp ice cream

Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes, or services transferred as extension activities)

Shrimp ice cream product

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